

REPORT

Introduction:

This project aimed to prepare an e-commerce dataset for machine learning by cleaning and preprocessing it. These tasks were to manage missing data, identify and remove outliers, create new data features, examine correlations, and label encoding categorical variables.

Dataset Description:

The data includes details about the e-commerce order, such as customer information, product information, payment channels, coupons used, referral sources, order status, quantity, unit price, and total price.

COLUMN	TYPE
OrderID	Text
Date	Date
CustomerId	Text
Product	categorical
Quality	Number
UnitPrice	Decimal
ShippingAddress	Text
PaymentMethod	categorical
OrderStatus	Text
TrackingNumber	Text
ItemsInCart	Number
CouponCode	Text
ReferralSource	categorical
TotalPrice	decimal

Missing values:

Coupon code: has some values missing

25.75% of the data was missing for the CouponCode column. The project guidelines specify that KNN imputation must be used for missing data greater than 20%, but KNN is not appropriate since CouponCode is a categorical variable. When the customers who didn't use a coupon were missing, they were replaced with "No Coupon" to maintain the business meaning of the data.

Number of outliers:

- No outliers in UnitPrice.
- No outliers in quantity

- No outliers in itemsincart
- The IQR method was used to identify 8 outliers in the TotalPrice column. These values were not revealed to be erroneous entries by the data entry staff, but were found to be legitimate high-value customer purchases after inspection. As a result, they were not altered but left in place.

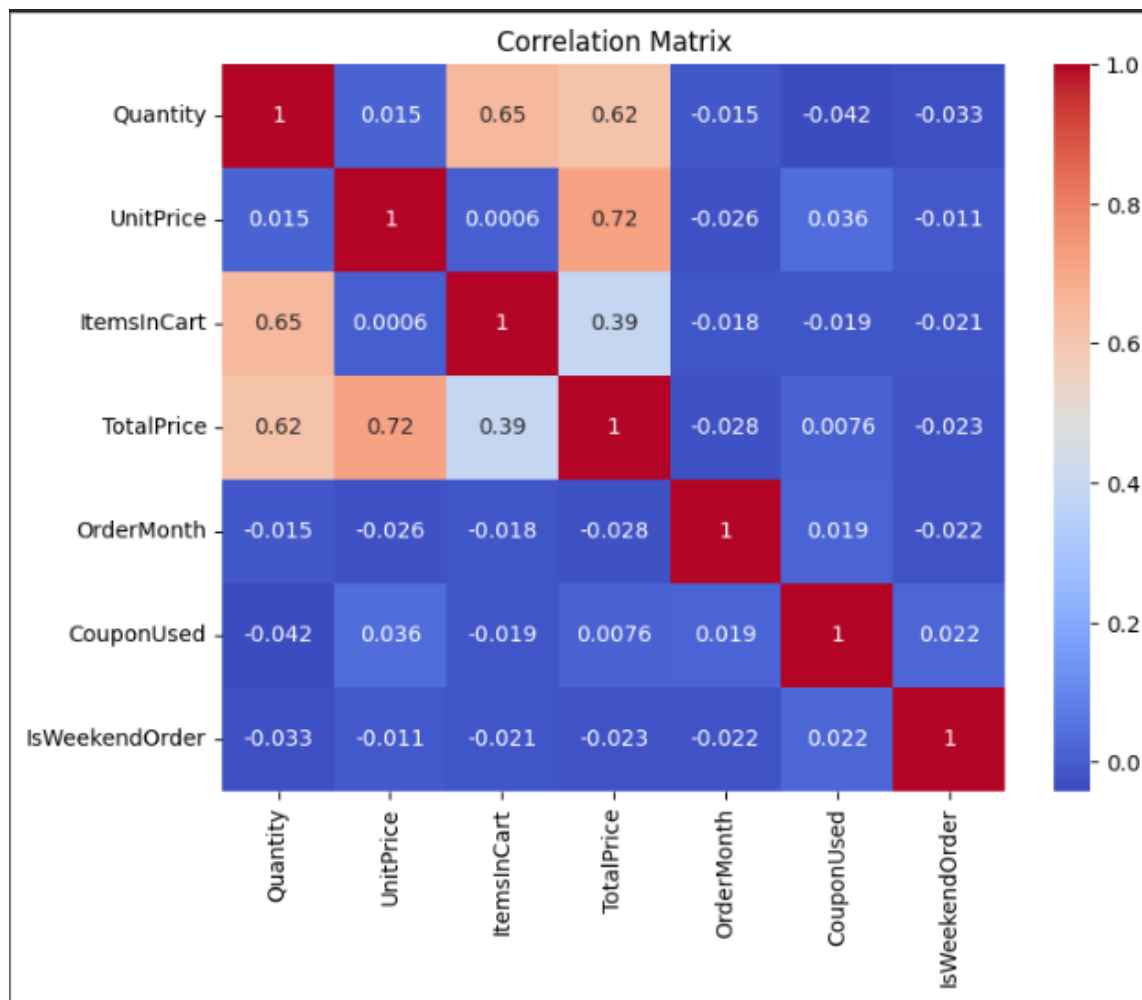
Feature engineering:

3 new features introduced:

- Order month: Extracted from the Date column to capture seasonal purchasing patterns.
- couponUsed: A binary feature indicating whether a coupon was applied.
- IsWeekendOrder: Indicates whether the order was placed on a weekend.

Correlation Analysis:

There were no pairs of variables with a correlation greater than 0.9; therefore, no additional variables were removed due to multicollinearity.



The correlation between the variables and multicollinearity:

A correlation matrix was created to explore the correlation between the numerical features of the dataset.

The analysis showed that:

- The TotalPrice is highly correlated with UnitPrice (0.72), as would be expected as the TotalPrice of an order is dependent on the UnitPrice of the products.
- The correlation between quantity and ItemsInCart is moderate positive (0.65), which means that customers who buy more items tend to have bigger carts.
- None of the pairs of features had a perfect or very high correlation (above 0.9) and therefore no further features were dropped due to multicollinearity.
- The correlation analysis showed that the engineered features (OrderMonth, CouponUsed, IsWeekendOrder) do not contain redundant information and they do provide unique information.

One Hot Encoding:

One Hot Encoding was applied to the categorical variables (Product, PaymentMethod, OrderStatus, and ReferralSource). To avoid the dummy variable trap, the parameter `drop_first=True` was used, and thus reducing redundancy among the generated features. The CouponCode column was not one-hot encoded because it had been transformed to the binary feature CouponUsed, indicating whether a coupon was used.

Conclusion:

The provided data was cleaned and properly formatted for machine learning. Appropriate methods of handling missing values, analysing outliers, creating new predictive features, checking for multicollinearity and encoding categorical variables were conducted. The final data set can be used for predictive modelling and subsequent data analysis.